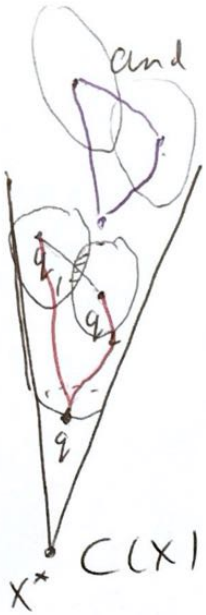
<u>Problem:</u> L' may have holonomy

Fix basepoint $q \in \partial B_1(x^*) \cap \mathrm{supp}(\psi_\rho)$



and paths joining q to q_i (can assume $\mathrm{supp}(\psi_\rho) \cap B_R(x^*)$ is connected)

use parallel transport to define constant sections σ_i' of L' over each $B(q_i, r_i)$ s.t. $|\sigma_i'| = 1$

We have finite collection of overlaps where $\dfrac{\sigma_i'}{\sigma_j'} \in S^1 \subset \mathbb{C}$

<u>Lemma</u> Given $\delta_1 > 0$ and $\{\lambda_1, \dots, \lambda_N\} \subset S^1$, $\exists K = K(\delta_1, N)$ & $m < K$ s.t. $|\lambda_i^m - 1| < \delta_1 \quad \forall i$

Replace line bundle L_k by L_k^m & scale the M_k by $\sqrt{m}$. Limit is still $C(X)$

Consider same $q_i \in C(X)$ and new points $q_i^{(k)} \in M_k$ s.t. $q_i^{(k)} \to q_i$

(<u>note</u>: original $q_i^{(k)}$ converge to $\sqrt{m}\, q_i$)

Repeat same construction. We get sections σ_i' over $B(q_i, r_i)$ s.t. $\left| \frac{\sigma_i'}{\sigma_j'} - 1 \right| < \delta_1 \quad \forall i,j$

Use partition of unity to glue σ_i' together to get section σ' of L' over $\mathrm{supp}(\psi_\rho) \cap B_R(x^*)$ s.t. $|\sigma' - \sigma_i'|, |\nabla(\sigma' - \sigma_i')| < \Psi(\delta_1 \mid n, v, R, \rho)$

$$(\nabla \sigma_i' = 0)$$

corresponding section σ of L satisfies

$$|\sigma - \sigma_i|, \ |\bar\partial \sigma| < e^{-\frac{1}{2} d(x^*, \cdot)^2} \Psi(\delta_1 \mid n, v, R, \rho)$$

$$|\sigma_i|^2 = e^{-d(x^*, \cdot)^2} \qquad < e^{-\frac{1}{2} d(x^*, \cdot)^2} \Psi(\rho, R^{-1} \mid n, v)$$

$$\Rightarrow \ |\sigma|^2 < 2 e^{-d(x^*, \cdot)^2}$$

if δ_1 chosen suitably small dep. on n, v, R, ρ

Let χ_R be cutoff function s.t.

$$\chi_R = \begin{cases} 1 & \text{in } B_{R-1}(x^*) \\ 0 & \text{outside of } B_R(x^*) \end{cases}$$

Let $\tilde\sigma = \psi_\rho \chi_R \sigma$, extended by zero outside of $\mathrm{supp}(\psi_\rho) \cap B_R(x^*)$

$$|\tilde\sigma|^2 < 2 e^{-d(x^*, \cdot)^2}$$

$$|\bar\partial \tilde\sigma| \leq \left(|\bar\partial \psi_\rho| + |\bar\partial \chi_R| \right) 2 e^{-\frac{1}{2} d(x^*, \cdot)^2}$$

$$+ \ \Psi(\rho, R^{-1} \mid n, v) \, e^{-\frac{1}{2} d(x^*, \cdot)^2}$$

$$\Rightarrow \quad \int_{C(X)} |\bar\partial \tilde\sigma|^2 < \Psi(\rho, R^{-1} | n, v)$$

We can find sections $\tilde\sigma_k$ of L_k converging to $\tilde\sigma$ s.t. $\int_{M_k} |\bar\partial \tilde\sigma_k|^2 < \Psi(\rho, R^{-1}, k^{-1} | n, v)$

Hörmander $\Rightarrow$ can solve $\bar\partial u_k = \bar\partial \tilde\sigma_k$

with $\| u_k \|_{L^2} < \Psi(\rho, R^{-1} | n, v)$ for $k \gg 1$

so $s_k = \tilde\sigma_k - u_k \in H^0(M_k, L_k)$

(recall we replaced L_k by L_k^m)

$$\int_{M_k} |\tilde\sigma_k|^2 < C(n, v)$$

$$\Rightarrow \quad \| s_k \|_{L^2} < C(n, v)$$

so $|s_k| + |\nabla s_k| < C(n, v)$

- On $B_{R-1}(x^*) \setminus B_\rho(S)$ $\quad \psi_\rho \chi_R = 1$

so $|\bar\partial \tilde\sigma| \leq \Psi(\rho, R^{-1} | n, v) e^{-\frac{1}{2} d(x^*, \cdot)^2}$

$$\Rightarrow \quad \overset{\bar\partial u_k}{|\bar\partial \tilde\sigma_k|} \leq \Psi(\rho, R^{-1} | n, v) e^{-\frac{1}{2} d(x^*, \cdot)^2} \text{ if } k \gg 1$$

Elliptic estimates $\Rightarrow \sup |u_k| \to 0$ on $B_{R-2}(x^*) \setminus B_{2\rho}(S)$

i.e. $s_k \xrightarrow{L^\infty} \sigma$ on $B_{R-2}(x^*) \setminus B_{2\rho}(S)$

using gradient estimate

$$\left| |s_k| - e^{-\frac{1}{2}d(p_k,\cdot)^2} \right| < \Psi(\rho, R^{-1}, k^{-1} | n, v) + \rho \cdot C(n,v)$$

on $B_{R-2}(x^*)$

$$< \delta \quad \text{if } \rho, R^{-1}, k^{-1} \ll 1$$

dep. on δ.

- Outside of $B_{R-3}(x^*)$,

$$\int_{M_k \setminus B_{R-3}(x^*)} |\tilde{\sigma}_k|^2 < \Psi(\rho, R^{-1} | n, v)$$

$$\Rightarrow \int_{M_k \setminus B_{R-3}(x^*)} |s_k|^2 < \Psi(\rho, R^{-1} | n, v)$$

by L^∞ estimate we have $|s_k| < \Psi(\rho, R^{-1} | n, v)$ on $M_k \setminus B_{R-2}(x^*)$ $\square$